CLASS 10 — SCIENCE · FORMULA SHEET

Chapter 6: **Control and Coordination** | Quick revision, one page

Nervous system

- Pathway: receptor → sensory neuron → CNS (spinal cord/brain) → motor neuron → effector (muscle/gland).
- Neuron: dendrite → cell body → axon → axon end → synapse (chemical transmission) → next neuron.
- Reflex arc: receptor → sensory neuron → spinal cord → motor neuron → effector (quick, involuntary).
- Receptors: eyes (light), ears (sound, balance), nose (smell), tongue (taste), skin (touch, temperature).

Human brain

- Forebrain: cerebrum (thinking, memory, voluntary actions, senses), hunger centre. Midbrain.
- Hindbrain: medulla (involuntary actions: blood pressure, salivation, vomiting), pons, cerebellum (posture, balance, precision of voluntary actions).
- Protection: skull (brain), vertebral column (spinal cord), cerebrospinal fluid.

Plant movements and hormones

- Tropism: directional growth in response to a stimulus. Phototropism (light), geotropism (gravity), hydrotropism (water), chemotropism (chemicals), thigmotropism (touch).
- Auxin: promotes cell elongation (shoot tip). Gibberellin: stem growth. Cytokinin: cell division (fruits, seeds). Abscisic acid: inhibits growth, causes wilting.
- Sensitive plant (Mimosa): non-directional response due to change in cell turgor.

Hormones in animals

- Thyroxine (thyroid): metabolism; needs iodine; deficiency causes goitre. Growth hormone (pituitary): growth; deficiency dwarfism, excess gigantism.
- Insulin (pancreas): lowers blood sugar; deficiency causes diabetes. Adrenaline (adrenal): fight-or-flight, raises heart rate and breathing.
- Testosterone (testes) and oestrogen (ovary): changes at puberty. Feedback mechanism regulates hormone release.

MASTERMIND